

Vacancy CTMC for Pollinator-Mediated Replacement

July 9, 2026

1 Purpose

This note describes the small continuous-time Markov chain (CTMC) used to compute the probability that an open plant vacancy is eventually filled by yeast or bacteria, allowing the local pollinator count to change before the vacancy is filled. The CTMC closure uses the original pollinator movement rates while the vacancy is open; there is no fitted pollinator-mixing parameter.

The implemented version is in `definitions.unified.wl`; the core helper is the function

`CTMCClosureReplacementProbabilityVectors.`

The model here should be read as a diagnostic or closure component, not as a replacement for the full three-species master equation.

2 State after a focal death

Consider a full local state

$$(Y, B, P) = (y, b, p).$$

If a bacteria-occupied site dies, the post-death plant state is

$$(y, b - 1),$$

and the vacancy may eventually be filled by yeast or bacteria. If a yeast-occupied site dies, the post-death plant state is

$$(y - 1, b).$$

During the small CTMC calculation, the post-death plant counts are held fixed and only the local pollinator count changes:

$$p \in \{0, 1, \dots, K\}, \quad K = P_{\max}.$$

The vacancy CTMC stops when a plant replacement occurs.

3 Transient and absorbing rates

Fix a post-death plant state (y, b) and a regional pool

$$(P_Y^R, P_B^R, P_\emptyset^R).$$

For each pollinator count p , define the following rates by evaluating the full-model transition rates at $(Y, B, P) = (y, b, p)$:

$$\begin{aligned} A_p &= f_2(y, b, p) + \mathbf{1}_{\{p < K\}} f_5(y, b, p), & \text{rate of replacement by yeast,} \\ C_p &= f_3(y, b, p) + \mathbf{1}_{\{p < K\}} f_6(y, b, p), & \text{rate of replacement by bacteria,} \\ U_p &= \mathbf{1}_{\{p < K\}} f_4(y, b, p), & \text{rate of pollinator immigration, } p \rightarrow p + 1, \\ D_p &= \mathbf{1}_{\{p > 0\}} f_1(y, b, p), & \text{rate of pollinator emigration, } p \rightarrow p - 1. \end{aligned}$$

The indicators impose the same finite $P_{\max}$ boundary as the full master equation. When $P_{\max}$ is chosen large enough, these indicators are merely a numerical truncation of an effectively infinite pollinator cloud.

4 Hitting probability equations

Let

$$q_p = \mathbb{P}(\text{vacancy is eventually filled by yeast} \mid P(0) = p)$$

for the post-death plant state (y, b) . First-step analysis gives

$$0 = A_p(1 - q_p) + C_p(0 - q_p) + U_p(q_{p+1} - q_p) + D_p(q_{p-1} - q_p).$$

Equivalently,

$$(A_p + C_p + U_p + D_p)q_p - U_p q_{p+1} - D_p q_{p-1} = A_p.$$

Writing $q = (q_0, \dots, q_K)^\top$, this is a tridiagonal linear system

$$Mq = A,$$

where

$$M_{p,p} = A_p + C_p + U_p + D_p, \quad M_{p,p+1} = -U_p, \quad M_{p,p-1} = -D_p,$$

with omitted entries outside $\{0, \dots, K\}$. The code solves this system with `LinearSolve`. The bacteria-filling probabilities can be solved with right-hand side $C = (C_0, \dots, C_K)^\top$, or, when absorption is certain, by

$$r_p = 1 - q_p.$$

For finite K , this is already a closed-form object in the sense that

$$q = M^{-1}A.$$

Because M is tridiagonal, the entries can also be written as finite continued fractions. In practice, the tridiagonal solve is the cleaner form for a closure.

5 Effective full-model hazards

Given a full stationary distribution $\pi(Y, B, P)$, the eventual-replacement diagnostic computes effective one-step composition-change hazards.

For a state (y, b, p) , the effective yeast-up hazard due to a bacteria death is

$$h_+(y, b, p) = db q_p^Y(y, b - 1),$$

where $q_p^Y(y, b-1)$ is the yeast-filling probability for the vacancy CTMC after a bacteria death.

Similarly, the effective yeast-down hazard due to a yeast death is

$$h_-(y, b, p) = d y q_p^B(y-1, b),$$

where $q_p^B(y-1, b)$ is the bacteria-filling probability for the vacancy CTMC after a yeast death.

The diagnostic plotted against the closures averages these hazards conditional on the current value of Y :

$$H_+(y_0) = \mathbb{E}_\pi[h_+(Y, B, P) \mid Y = y_0], \quad H_-(y_0) = \mathbb{E}_\pi[h_-(Y, B, P) \mid Y = y_0].$$

Figure 1 shows this comparison for the $P_R = 1$ coexistence calculation in the exploration notebook. The full-model curves use the eventual vacancy-replacement CTMC described above, while the two dashed curves are the mean-field and Poisson-averaged closures.

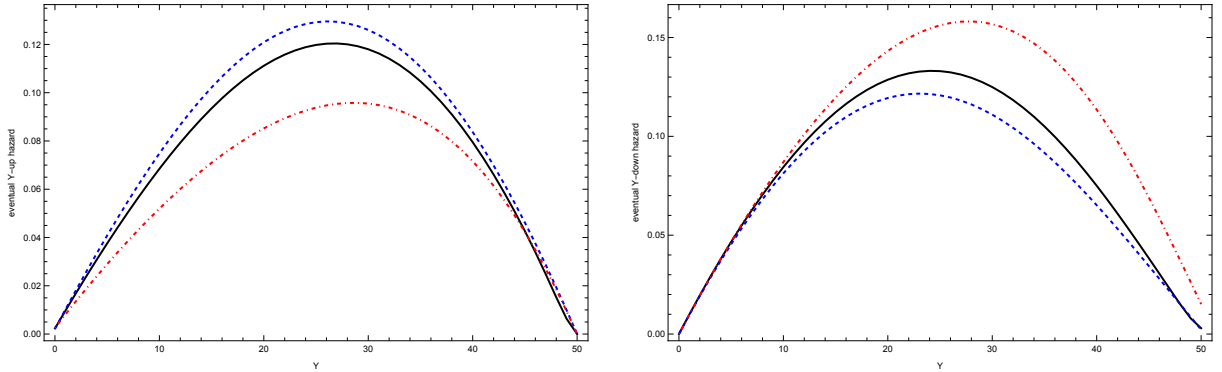


Figure 1: Full-model eventual vacancy-replacement hazards compared with the mean-field (MF) and Poisson-averaged (Pois) closure hazards for the $P_R = 1$ coexistence calculation. The left panel shows effective yeast-up hazards and the right panel shows effective yeast-down hazards, both conditional on the current value of Y . Black solid curves are the full eventual-vacancy hazards; blue dashed curves are MF; red dot-dashed curves are Pois.

6 Fixed-timescale CTMC closure

The reduced CTMC closure uses the same hitting problem as above, then averages over the local pollinator cloud expected at the instant a plant dies. For a closed reduced state with $Y = y$ and $B = n - y$, let

$$w_p(y) = \mathbb{P}(P = p \mid Y = y)$$

denote the truncated, normalized Poisson weights used by the code. The mean of this cloud is the usual conditional pollinator mean from the reduced model.

After a bacteria death, the post-death plant state is $(y, n - y - 1)$. Define

$$Q_{B \rightarrow \emptyset}^Y(y) = \sum_{p=0}^K w_p(y) q_p^Y(y, n - y - 1),$$

where $q_p^Y(y, n - y - 1)$ is the yeast-filling probability from the CTMC. After a yeast death, define analogously

$$Q_{Y \rightarrow \emptyset}^B(y) = \sum_{p=0}^K w_p(y) q_p^B(y - 1, n - y),$$

where q_p^B is the bacteria-filling probability.

The CTMC closure then gives the one-dimensional transition hazards

$$u_y = d(n - y)Q_{B \rightarrow \emptyset}^Y(y), \quad v_y = dyQ_{Y \rightarrow \emptyset}^B(y),$$

where u_y is the yeast-up hazard $y \rightarrow y + 1$ and v_y is the yeast-down hazard $y \rightarrow y - 1$.

7 Relation to the simpler closures

The CTMC closure differs from the two simpler closures in when the pollinator cloud is averaged and whether it can move while the vacancy is open.

7.1 Poisson-hazard closure

If the pollinator count is held fixed after the plant death, then the hitting equations decouple:

$$q_p = \frac{A_p}{A_p + C_p}.$$

If the local pollinator count is treated as a Poisson random variable

$$P \sim \text{Pois}(\lambda),$$

then the replacement probability is

$$\mathbb{E} \left[\frac{A_P}{A_P + C_P} \right].$$

This is the Poisson-averaged closure: first compute the instantaneous replacement probability at a fixed pollinator count, then average that nonlinear quantity over the pollinator cloud.

7.2 Mean-field closure

If, instead, the pollinator cloud is collapsed to its mean, then

$$P = \lambda$$

and

$$q^{\text{MF}} = \frac{A_\lambda}{A_\lambda + C_\lambda}.$$

This is the mean-field closure: replace the random pollinator count by its conditional mean before computing the replacement probability.

8 Summary of the closures

The three reduced closures can be summarized as

$$\begin{aligned}
 P = \lambda & \implies \text{mean-field closure,} \\
 P \sim \text{Pois}(\lambda), P \text{ fixed during replacement} & \implies \text{Poisson-hazard closure,} \\
 P \sim \text{Pois}(\lambda), P \text{ mobile during replacement} & \implies \text{fixed-timescale CTMC closure.}
 \end{aligned}$$

The CTMC closure is therefore a useful intermediate object: it keeps the nonlinear replacement race, starts from a Poisson local cloud, and allows that cloud to move before the vacancy is filled.

9 Illustrative CTMC figures

Figure 2 shows the distribution of the local pollinator count at the instant the vacancy is filled,

$$\mathbb{P}(P_{\text{birth}} = p \mid P(0) = p_0),$$

for three initial pollinator counts. The probabilities sum over yeast and bacteria replacement events. Columns are arranged from slower to faster pollinator movement: no pollinator change while the vacancy is open, the CTMC closure, and the Poisson cloud used by the Poisson-hazard closure. The CTMC column is intermediate: it is not locked to p_0 , but it still remembers the initial pollinator count.

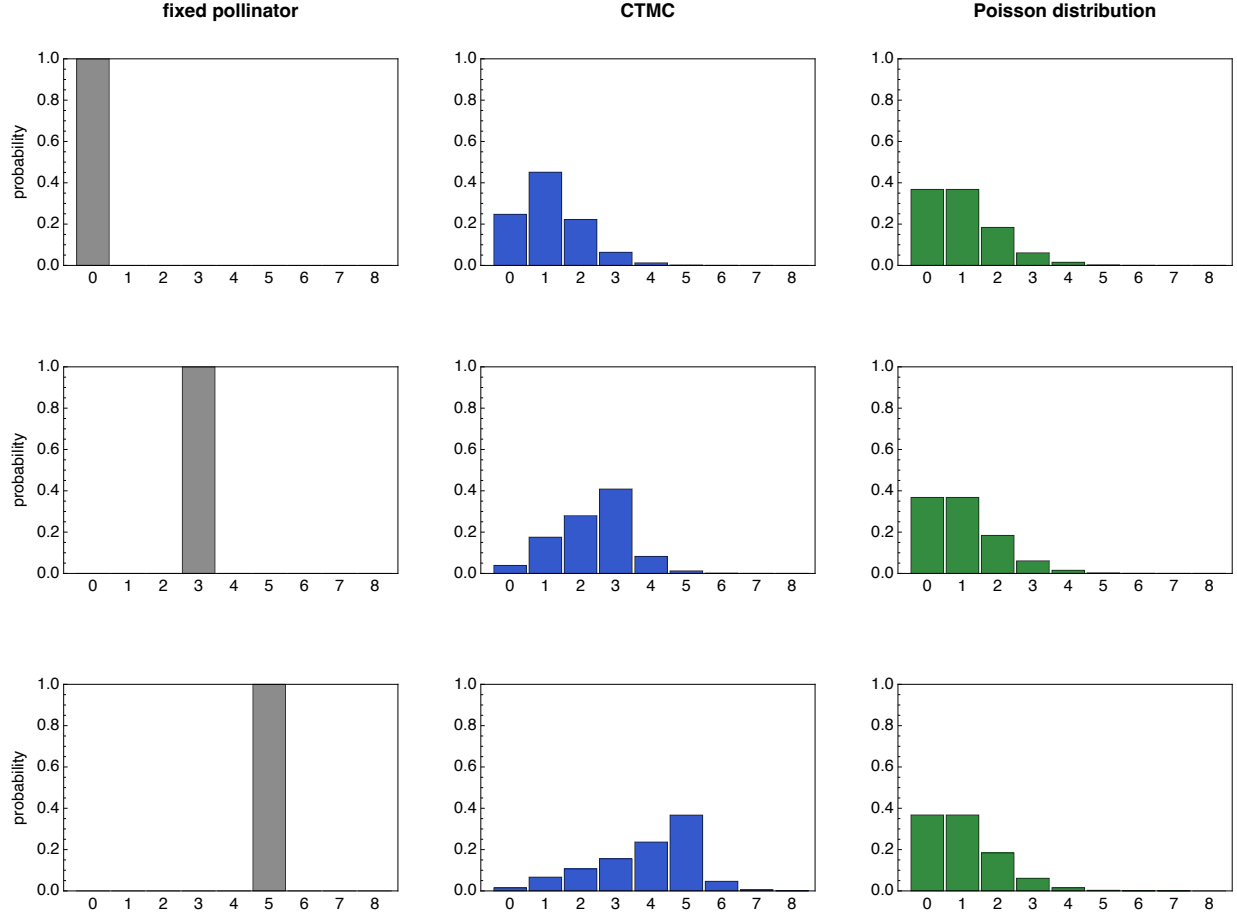


Figure 2: Distribution of the local pollinator count p at the replacement event for initial pollinator counts $p_0 = 0, 3, 5$ from top to bottom. The columns show the fixed pollinator endpoint, the fixed-timescale CTMC closure, and the Poisson distribution endpoint.

Figure 3 shows the corresponding probability that a death-created opening is eventually filled by yeast as a function of the current closed-state yeast abundance. For a given y , the plotted value is

$$\frac{(n - y)Q_{B \rightarrow \emptyset}^Y(y) + y(1 - Q_{Y \rightarrow \emptyset}^B(y))}{n},$$

the death-event-weighted probability that the replacement is yeast.

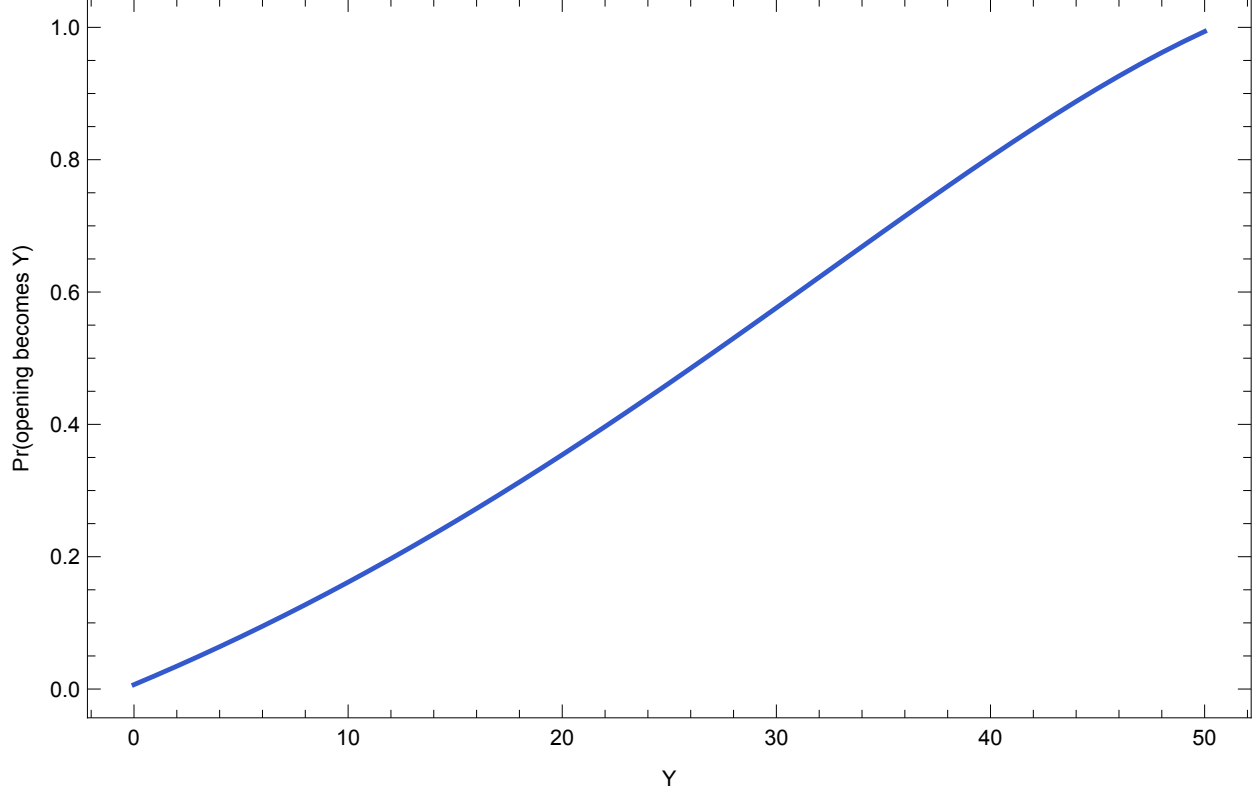


Figure 3: Probability that a death-created opening is eventually filled by yeast as a function of the current yeast abundance Y under the fixed-timescale CTMC closure.

10 Truncation note

The analytic closure picture is cleanest for an infinite pollinator cloud. The implemented full model and diagnostic use a finite cutoff $P_{\max} = K$. To make the finite calculation comparable to the infinite Poisson closure, K should be large enough that both the ordinary Poisson tail and the rate-weighted tail are negligible over the relevant range of Y :

$$\mathbb{P}(\text{Pois}(\lambda_{\max}) > K) \ll 1, \quad \frac{\mathbb{E}[P \mathbf{1}_{\{P > K\}}]}{\mathbb{E}[P]} \ll 1.$$

For $P \sim \text{Pois}(\lambda)$,

$$\frac{\mathbb{E}[P \mathbf{1}_{\{P > K\}}]}{\lambda} = \mathbb{P}(\text{Pois}(\lambda) \geq K).$$

It is also worth checking directly in the full stationary distribution that

$$\pi(P = K)$$

is negligible; otherwise the cutoff is shaping the full model rather than just truncating an effectively infinite cloud.